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This document describes a data format, the NDIC Data Distribution Interface (DDR), used to provide information from floating vehicles (travel times, speed, traffic conditions and queue length, number of vehicles, types) at a predefined set of locations. Separated by vehicle type (car, truck). The data format is not backward compatible with the original format, introduces a hierarchical structure to the elements, renames them and modifies the meaning.

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format and other schemas (RELAX NG)
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format

description of data structure for transferring some kind of information

protocol

description of data transfer process and rules

NDIC

national traffic information centre

DDR

data distribution interface, a proprietary format used by NDIC

(W3C) XML schema

language, which allows definition of rules for a XML document structure.

RELAX NG schema

human readable language, which allows definition of rules for a XML document structure.

XSD

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

PLS

a set (collection) of predefined locations, usually used to describe a location by a reference from a dynamic to a static publication.

uri

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CONCEPTS

This chapter describes the basic concepts used in this format. Including a description of the elements and their meaning. Detailed definitions are also given in the diagrams. In addition, this chapter provides short examples from actual XML documents. For more examples, see [Appendix A: Message examples](#).

The described format allows the provision of information extracted from FCD. These are travel times, traffic flow speeds, congestion levels and estimates of the length and position of queues on a set of predefined routes, differentiated by vehicle type into information for trucks and for cars.

2.1 Parts of the XML document

The XML document builds on the following elements and parts:

- *DOC* (document root)
- *INF* (metadata)
- *FCD* (container for state data on a segment)
- *SEGMENT* (statuses per segment)

Unlike the original version of x-format:en-ndic_ddr-fcd-v1.0, this introduces structure to the description, renames some elements and attributes, and modifies their meaning.

2.1.1 The document root (*DOC*)

The *DOC* element represents a single publication (a single instance of the transmitted data).

The *xmlns* attribute refers to the schema version used.

The *created* attribute indicates the time at which the publication, as a whole, was created in a dateTime format.

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <DOC xmlns="x-format:cz-ndic_ddr-fcd-v2.0" created="2023-01-23T18:28:18+01:00">
```

2.1.2 Metadata (*INF*)

The *INF* element describes meta-information about the entire publication, specifically in the DAT/D2PLS sub-element it provides a reference to the DATEX II publication containing a description of the positions of the segments.

```
1 <INF>
2   <DAT>
3     <D2PLS uri="x-source:cz-ndic_d2-pls-fcd"/>
4   </DAT>
5 </INF>
```

2.1.3 Container for status data (*FCD*)

The individual segment statuses are placed in a common *FCD* parent element. This element contains information about those segments, out of the total number of segments, on which vehicle probes are or **recently** have been present.

The recent time is now set to T-5 minutes, T being the time of creation of the given publication.

Segments on which vehicle probes are not or have not been present in **recent** time are **not** listed.

2.1.4 Status of segment (*SEGMENT*)

The traffic situation on a segment is described by the sub-elements of the *SEGMENT* element:

- element *ALL* contains values/statuses without differentiation of vehicle types, always present.
- The elements *CAR* and *TRUCK* contain values/statuses for passenger vehicles and trucks respectively.

Elements *SEGMENT* are distinguished (by content only, not by attribute) into:

- segments with **mixed traffic flow**, where it does not make sense to evaluate traffic parameters separately by vehicle type, i.e. vehicles moves in a mixed traffic or in single lane traffic. For these segments only the element *ALL* is given.
- Segments with **separate traffic flow** where it makes sense to evaluate traffic parameters separately by vehicle type, i.e. vehicles move in separated traffic streams, in multiple lanes, so separated queues of vehicles by vehicle type may occur. For these segments, the element *ALL* and one or both of the elements *CAR* and *TRUCK* are provided depending on whether the vehicles of the type were present on the segment at the time of evaluation.

Segment identity and version and time

The identity of the state on a segment (*SEGMENT*) is uniquely determined by its position (given by the position identifier from *D2PLS* and its version) and the time of the calculation of the values.

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```
<SEGMENT id="TS42739T17195" version="1" last_updated="2023-01-23T18:26:06+01:00">
```

The *id* attribute gives the identity of the segment, referring to the publication of predefined positions given in the metadata see *Metadata (INF)* to a record with the same *id* (and version).

Note: Position descriptions may evolve over time, so multiple entries with the same identifier may appear in a predefined locations publication, differing in their version.

The *version* attribute indicates the version of a given location description, and together with the value in *id* forms a unique reference to a set of predefined locations.

The *last_updated* attribute indicates the time for which the data is calculated, i.e. the last time values were calculated (and updated) for *SEGMENT*. The update takes place regularly at an interval of 1 minute. At that time, the routes of all vehicles with respect to travelled sections (parts of segments) of the monitored network are evaluated. The last update of the segment status occurs when the last vehicle leaves the segment.

Segments whose last update took place more than 4-5 minutes ago are not included in the overall overview (they are not up-to-date).

Values without vehicle type distinction (*ALL*)

The element *ALL* contains the calculated values on the segment for all vehicles without type distinction and an element describing the queue on the segment (*CONGESTION*) if the real queue is present.

```
<ALL count="1" speed="41" travel_time="30" delay="5" freeflow_speed="50"
freeflow_time="25" reliability="0.500" traffic_level="2"/>
```

Note: The following attributes are always for all vehicles on a given section without type distinction.

The attribute *count* indicates the number of vehicles used for the calculation of the values (without type distinction).

The *speed* attribute gives the current speed of the traffic stream [km/h]. Calculated as the average speed of the vehicles that occur on the segment.

The *travel_time* attribute gives the current travel time [s]. The current travel time information is calculated from the “segment length/average speed on the segment”.

The *delay* attribute gives the delay [s]. The delay information is calculated from “free segment travel time - current travel time”.

The *freeflow_speed* attribute gives the free passage speed [km/h]. Calculated as historical values of the free-flow speed aggregated over all periods.

The *freeflow_time* attribute indicates the free passage time [s]. Calculated as “segment length”/freeflow_speed.

Note: The length of the entire segment can be calculated from the values of *freeflow_speed* and *freeflow_time* of the *ALL* element, which is always present, or found from publishing a set of predefined positions.

Attribute *reliability* indicates the reliability of the data, which characterizes the quality of the data sample [%]. It has values <0-1> where 1 means 100% reliable data. The value depends only on the number of vehicles on the segment, the base value is 50% for 1 vehicle on the segment, with more vehicles on the segment is the value asymptotically increasing to 100%. The reliability value can therefore always be calculated from knowing the number of vehicles per segment.

The *traffic_level* attribute indicates the level of traffic according to the custom in the country [-]. Here it is calculated as the % difference between the current traffic flow speed and the traffic flow speed on the free segment.

Table 1: Calculation in the definition of traffic stages in the FCD publication

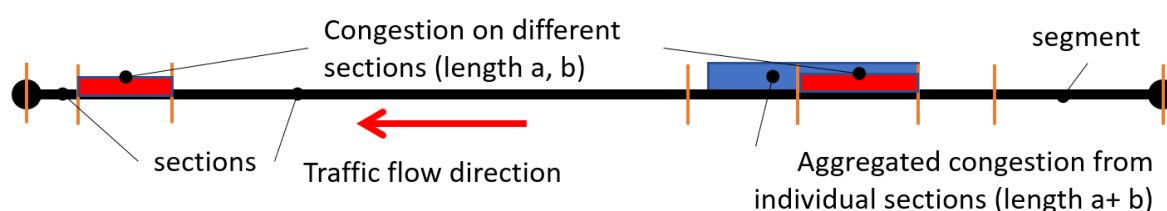
Degree	Description	% of difference
1	single vehicle operation only, driving is smooth	<85 %, 100 %)
2	small groups of vehicles, smooth running, passing intersections without problems	<75 %, 85 %)
3	vehicle streams are forming, traffic is smooth but speed is below the maximum speed limit	<50 %, 75 %)
4	vehicle queues are forming, traffic is not flowing, average speed is significantly reduced, passing the intersections is impaired	<20 %, 50 %)
5	traffic collapse - vehicles are stationary on the roads or moving very slowly in convoys, average speed is very low	<0 %, 20 %)

Congestion on a segment (*CONGESTION*)

Segments are internally divided into smaller parts (sections of the Global Network, GN road network). The presence of congestion is evaluated with respect to GN sections as a decrease in the travelled speed of this section **below 30 % of the free-flow speed**.

Note: The distinction is made if congestion is generated by cars or trucks. The *CONGESTION* element is therefore present in the element *ALL* with attributes computed from all vehicles on congested sections, and in the element *TRUCK* or *CAR* computed only from given type of vehicles.

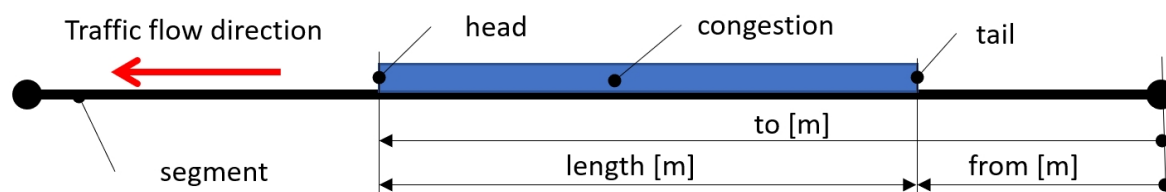
The *CONGESTION* element contains aggregated values from individual congested sections of the segment. If none of the sections of the segment were congested, the *CONGESTION* element is not present. The element is present only once per segment, the values of its attributes are adapted to this (for more, see the description of the attributes and the image below).



Note: A congestion may be present on a segment even if the current speed on the whole segment is comparable to or higher than the free flow segment speed, for example, a congestion on a short section before an intersection.

```
<CONGESTION from="98" length="171"/>
```

The *from* attribute indicates the start of congestion [m] measured from the start of the segment in the direction of the traffic flow. In the case of multiple split congestions on a segment, for example, a congestion detected on GN sections 1 and 3, the *from* attribute only gives information about the 1st congestion



The *length* attribute gives the aggregated length of the traffic congestion for all sections of the segment [m]. i.e. the sum of the lengths of the **congested** Global Network sections (even non-contiguous) lying on the segment.

Values for passenger vehicles (*CAR*)

The *CAR* element contains calculated values, for the segment with separate traffic flow, for passenger cars only (vehicles of categories M1, M2, M3, N1). It contains the number of passenger vehicles (*count*), the speed (*speed*) and the free flow speed (*freeflow_speed*).

Element describing the passenger vehicles queue on the segment (*CONGESTION*) if the real queue is present.

```
<CAR count="3" speed="128" freeflow_speed="130"/>
```

Values for trucks (*TRUCK*)

The *TRUCK* element contains the calculated values, for the segment with separate traffic flow, for trucks only (vehicles of category N2, N3). It contains the number of trucks (*count*), the speed (*speed*) and the free flow speed (*freeflow_speed*).

Element describing the trucks queue on the segment (*CONGESTION*) if the real queue is present.

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```
<TRUCK count="1" speed="90" freeflow_speed="80"/>
```

LOCATION DESCRIPTION

The publication of dynamic data described by this format, does not contain location descriptions, but refers to the publication of a set of predefined locations in DATEX II format. The publication with a set of predefined locations describes the locations of segments using ALERT-C, Open LR, Global Network sections and mileage methods.

For historical reasons, the location description is also included in the SEGMENT identifier, i.e. can be derived from the shape of its identifier. This interpretation is no longer maintained and may not always be correct. The concept is based on describing the location of a segment in relation to the Czech ALERT-C location table. The interpretation of the *id* of a SEGMENT is as follows: TS01409T04122

- TS ... the first 2 characters stand for traffic *SEGMENT* (traffic segment)
- 01409 ... the next 5 characters are the numeric code of the so-called primary location from the ALERT-C table,
- T ... location code separator
- 04122 ... the last 5 characters are the numeric code of the so-called secondary location from the ALERT-C table,

The primary location is the head of the problem, the point where, in the direction of the traffic flow, the driver leaves the location, usually the location of the accident site. The secondary location is the location where the driver enters the location, usually the tail of the queue (increasing upstream from the primary location).

APPENDIX A: MESSAGE EXAMPLES

4.1 fcd-data-v2-01.xml - sample of possible combinations of statuses

The message contains several different combinations of statuses on segments depending on whether the traffic stream is mixed or separated and whether congestion is present on the segment.

```
<?xml version="1.0" encoding="UTF-8"?>
<DOC xmlns="x-format:cz-ndic_ddr-fcd-v2.0" created="2023-01-23T18:28:18+01:00">
  <INF>
    <DAT>
      <D2PLS uri="x-source:cz-ndic_d2-pls-fcd"/>
    </DAT>
  </INF>
  <FCD>
    <!-- sdružený dopravní proud bez kongesce -->
    <SEGMENT id="TS42739T17195" version="1" last_updated="2023-01-23T18:26:06+01:00">
      <ALL count="1" speed="41" travel_time="30" delay="5" freeflow_speed="50"
      ↪ freeflow_time="25" reliability="0.500" traffic_level="2"/>
    </SEGMENT>
    <!-- sdružený dopravní proud s kongescí -->
    <SEGMENT id="TS17576T42659" version="1" last_updated="2023-01-23T18:24:06+01:00">
      <ALL count="1" speed="5" travel_time="193" delay="175" freeflow_speed="50"
      ↪ freeflow_time="18" reliability="0.500" traffic_level="5">
        <CONGESTION from="98" length="171"/>
      </ALL>
    </SEGMENT>
    <!-- dělený dopravní proud s osobními vozidly a nákladními vozidly a s kongescí
    ↪ nákladních vozidel-->
    <SEGMENT id="TS22988T22989" version="1" last_updated="2023-01-23T18:28:05+01:00">
      <ALL count="3" speed="17" travel_time="484" delay="383" freeflow_speed="82"
      ↪ freeflow_time="101" reliability="0.750" traffic_level="4">
        <CONGESTION from="0" length="1936"/>
      </ALL>
      <CAR count="1" speed="6" freeflow_speed="81"/>
      <TRUCK count="2" speed="24" freeflow_speed="79">
        <CONGESTION from="0" length="1936"/>
      </TRUCK>
    </SEGMENT>
    <!-- dělený dopravní proud s osobními vozidly a nákladními vozidly a s kongescí
    ↪ nákladních i osobních vozidel-->
    <SEGMENT id="TS22989T22988" version="1" last_updated="2023-01-23T18:28:05+01:00">
      <ALL count="5" speed="37" travel_time="78" delay="52" freeflow_speed="114"
      ↪ freeflow_time="26" reliability="0.900" traffic_level="4">
```

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```

    <CONGESTION from="220" length="232"/>
  </ALL>
  <CAR count="2" speed="35" freeflow_speed="114">
    <CONGESTION from="320" length="112"/>
  </CAR>
  <TRUCK count="3" speed="43" freeflow_speed="80">
    <CONGESTION from="220" length="225"/>
  </TRUCK>
</SEGMENT>
<!-- dělený dopravní proud s osobními vozidly a nákladními vozidly -->
<SEGMENT id="TS01225T25680" version="1" last_updated="2023-01-23T18:28:05+01:00">
  <ALL count="5" speed="112" travel_time="105" delay="14" freeflow_speed="130"
↪ freeflow_time="91" reliability="0.900" traffic_level="1"/>
  <CAR count="3" speed="128" freeflow_speed="130"/>
  <TRUCK count="2" speed="84" freeflow_speed="80"/>
</SEGMENT>
<!-- dělený dopravní proud pouze s nákladními vozidly -->
<SEGMENT id="TS01203T25650" version="2" last_updated="2022-04-06T10:38:13+02:00">
  <ALL count="1" speed="90" travel_time="105" delay="33" freeflow_speed="130"
↪ freeflow_time="72" reliability="0.500" traffic_level="3"/>
  <TRUCK count="1" speed="90" freeflow_speed="80"/>
</SEGMENT>
<!-- dělený dopravní proud pouze s osobními vozidly -->
<SEGMENT id="TS01203T25655" version="1" last_updated="2023-01-23T18:28:05+01:00">
  <ALL count="2" speed="80" travel_time="100" delay="37" freeflow_speed="130"
↪ freeflow_time="63" reliability="0.750" traffic_level="3"/>
  <CAR count="2" speed="80" freeflow_speed="130"/>
</SEGMENT>
</FCD>
</DOC>

```

APPENDIX B: REPRESENTATIVE XML SCHEMA: RELAX NG

The RELAX NG schema in a compact format is used to describe the XML structure. At the same time, the traditional W3C XML Schema 1.0, created by conversion from RELAX NG, is also available.

5.1 RELAX NG Compact Syntax Schema (RNC)

RNC schema describes expected structures of XML documents.

Root element is introduced by *start =*

All other *<SomeName> =* elements define named patterns.

In our schema there are following sections:

- prolog (namespace declaration)
- includes (not used here)
- root element definition (*start =*)
- patterns for elements
- patterns for attributes
- patterns for data types

Comments *##* specify more details about meaning of particular parts (in **Czech** language).

5.1.1 *schema-cs.rnc* - root of the schema

```
default namespace = "x-format:cz-ndic_dds-fcd-v2.0"

start =

## kořenový element publikace
element DOC {

    ## čas vytvoření dokumentu
    attribute created { xsd:dateTime },

    ## kontextové informace k publikaci
    element INF {

        ## informace o souvisejících datových sadách
        element DAT {

            ## DATEX II sada předdefinovaných míst
            element D2PLS {
```

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```

    ## uri zdroje (implicitně nebo explicitně zahrnuje i verzi)
    attribute uri { xsd:NMTOKEN }
  }
},

## kontejner pro informace z jednotlivých segmentů
element FCD {

  SEGMENT+
}

SEGMENT =
  ## informace pro jeden segment
  element SEGMENT {

    ## identifikátor segmentu ze sady předdefinovaných míst
    attribute id { xsd:NCName },

    ## verze popisu polohy segmentu s daným id
    attribute version { xsd:integer },

    ## čas, pro který jsou data vypočtena [dateTime]
    ##
    ## poslední okamžik kdy byly pro segment vypočteny (aktualizovány) hodnoty.
    ## K aktualizaci dochází pravidelně v intervalu 1 minuty, kdy se vyhodnocují
    ↪ trasy všech vozidel
    ## na sledované síti a jimi projeté úseky a vyhodnocují se hodnoty k takto
    ↪ projetým úsekům.
    ## K poslední aktualizaci stavu segmentu dojde, když jej opustí poslední
    ↪ vozidlo.
    ##
    ## Segmenty jejichž poslední aktualizace proběhla před více než 5 minutami nejsou
    ↪ do celkového přehledu poskytnuty (nejsou aktuální).
    attribute last_updated { xsd:dateTime },

    ALL,

    CAR?,

    TRUCK?
  }+

ALL =

  ## hodnoty bez rozlišení typů vozidel (sdružený dopravní proud)
  element ALL {

    ## počet vozidel použitých pro výpočet hodnot [-]
    attribute count { xsd:integer },

    ## aktuální rychlost dopravního proudu [km/h]
    ##
    ## průměrná rychlost vozidel které se na segmentu vyskytují.

```

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```

attribute speed { xsd:integer },

## aktuální dojezdová doba [s]
##
## informace o aktuálním čase průjezdu vypočtená z „délka segmentu/průměrná_
↪ rychlost na segmentu“
attribute travel_time { xsd:integer },

## zpoždění [s]
##
## informace o zdržení vypočtená z jako „čas průjezdu volného segmentu - aktuální_
↪ čas průjezdu“
attribute delay { xsd:integer },

## stupeň dopravy dle zvyklostí v ČR [-]
##
## https://cs.wikipedia.org/wiki/Stupe%C5%88_provozu;
## Zde vypočítané z % rozdílu aktuální rychlosti dopravního
## proudu a rychlosti dopravního proudu na volném segmentu.
attribute traffic_level {

## stupeň 1: provoz pouze jednotlivých vozidel, jízda je plynulá
## rozmezí proc. rozdílu rychlostí < 85 %, 100 % )
    "1" |

## stupeň 2: provoz malých skupinek vozidel, jízda je plynulá, odbavování na_
↪ křižovatkách bez problémů
## rozmezí proc. rozdílu rychlostí < 75 %, 85 % )
    "2" |

## stupeň 3: tvoří se proudy vozidel, provoz je plynulý, avšak rychlost nižší než_
↪ maximální povolená
## rozmezí proc. rozdílu rychlostí < 50 %, 75 % )
    "3" |

## stupeň 4: tvoří se kolony vozidel, provoz není plynulý, průměrná rychlost je_
↪ výrazně snižena, průjezd křižovatkami je narušen
## rozmezí proc. rozdílu rychlostí < 20 %, 50 % )
    "4" |

## stupeň 5: dopravní kolaps - vozidla na komunikacích stojí nebo v kolonách jen_
↪ velmi pomalu popojíždějí, průměrná rychlost je velmi malá
## rozmezí proc. rozdílu rychlostí < 0 %, 20 % )
    "5"
},

## rychlost volného průjezdu [km/h]
##
## historická hodnota rychlosti volného průjezdu
## agregovaná přes všechna období.
attribute freeflow_speed { xsd:integer },

## doba volného průjezdu [s]
##
## historická hodnota vypočtena jako délka segmentu/rychlost volného průjezdu
attribute freeflow_time { xsd:integer },

```

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```

    ## spolehlivost dat, která charakterizuje kvalitu datového vzorku [%]
    ##
    ## Nabývá hodnot <0-1> kdy 1 znamená 100% spolehlivá data.
    ## Závislá pouze na počtu vozidel na segmentu, základní hodnota 50 % pro 1_
    ↪ vozidlo
    ## na segmentu, s více vozidly na segmentu asymptoticky stoupá ke 100 %
    attribute reliability { xsd:decimal },

    CONGESTION?
}

```

CAR =

```

## hodnoty pro osobní vozidla
element CAR {

    ## počet vozidel použitých pro výpočet hodnot [-]
    attribute count { xsd:integer },

    ## aktuální rychlost dopravního proudu [km/h]
    ##
    ## průměrná rychlost vozidel které se na segmentu vyskytují.
    attribute speed { xsd:integer },

    ## rychlost volného průjezdu [km/h]
    ##
    ## historická hodnota rychlosti volného průjezdu
    ## agregovaná přes všechna období.
    attribute freeflow_speed { xsd:integer },

    CONGESTION?
}

```

TRUCK =

```

## hodnoty pro nákladní vozidla
element TRUCK {

    ## počet nákladních použitých pro výpočet hodnot
    attribute count { xsd:integer },

    ## aktuální rychlost dopravního proudu [km/h]
    ##
    ## průměrná rychlost vozidel které se na segmentu vyskytují.
    attribute speed { xsd:integer },

    ## rychlost volného průjezdu [km/h]
    ##
    ## historická hodnota rychlosti volného průjezdu
    ## agregovaná přes všechna období.
    attribute freeflow_speed { xsd:integer },

    CONGESTION?
}

```

CONGESTION =

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```

## Popis konkrétní kolony
##
## Segmenty jsou interně děleny na menší části (úseky GN).
## Na kterékoliv z těchto interních částí vzniká kolona,
## pokud na ní rychlost klesne pod 25% rychlosti volného průjezdu.
##
## V současné chvíli se nerozlišuje je-li kolona generována osobními
## či nákladními vozidly, do výpočtu tedy vstupují všechny úseky segmentu
## na kterých vozidlo či vozidla zpomalila na 30 % rychlosti volného průjezdu.
element CONGESTION {

    ## začátek kolony [m] měřeno od počátku segmentu ve směru dopravního proudu
    ##
    ## v případě více rozdělených kolon na segmentu, například kolona detekovaná
    ## na 1. a 3. úseku GN atribut from udává pouze informaci o 1. koloně
    attribute from { xsd:integer },

    ## agregovaná délka kolony na všech úsecích segmentu [m]
    ##
    ## resp. délka úseku GN na kterém byla kolona detekována.
    ## v případě více rozdělených kolon na segmentu atribut length
    ## udává sloučenou délku všech úseků segmentu s detekovanou kolonou
    attribute length { xsd:integer }

}

```

CHANGELOG

version 2.0

- Rewrite of CONGESTION tag, where formerly was outside of ALL, CAR and TRUCK tags, now is inside allowing to have different values for different types of traffic flows.

version 1.1.0

- Complete rewrite of the specification. Formerly based on a tmc tag holding in as attributes all values
- the information is now structured into SEGMENTS, with id, version and update time and information contained within the tag is furthermore separated in to tags - ALL with traffic parameters for non-divided traffic flow - CAR and TRUCK for information about separated traffic flows - CONGESTION for congestion length information

version 1.0.0

- Initial format and documentation

ANNEX C: ALTERNATIVE XML SCHEMA: W3C XML SCHEMA 1.0 (XSD)

W3C XML Schema 1.0 is probably the most popular format for describing XML structures today.

The schema below was created by transforming from the RNC schema and should show all the rules from the RNC. In the (unlikely) event that there are differences, consider the RELAX NG RNC as the reference version.

Note: Schema comments / definitions are in the **Czech** language.

7.1 *schema-cs* - schema root

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema" elementFormDefault="qualified"
  ↪targetNamespace="x-format:cz-ndic_ddr-fcd-v2.0" xmlns:c="x-format:cz-ndic_ddr-fcd-
  ↪v2.0">
  <xs:element name="DOC">
    <xs:annotation>
      <xs:documentation>kořenový element publikace</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="c:INF"/>
        <xs:element ref="c:FCD"/>
      </xs:sequence>
      <xs:attribute name="created" use="required" type="xs:dateTime">
        <xs:annotation>
          <xs:documentation>čas vytvoření dokumentu</xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="INF">
    <xs:annotation>
      <xs:documentation>kontextové informace k publikaci</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="c:DAT"/>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
  <xs:element name="DAT">
```

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```

<xs:annotation>
  <xs:documentation>informace o souvisejících datových sadách</xs:documentation>
</xs:annotation>
<xs:complexType>
  <xs:sequence>
    <xs:element ref="c:D2PLS"/>
  </xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="D2PLS">
  <xs:annotation>
    <xs:documentation>DATEX II sada předdefinovaných míst</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute name="uri" use="required" type="xs:NMTOKEN">
      <xs:annotation>
        <xs:documentation>uri zdroje (implicitně nebo explicitně zahrnuje i verzi)</
↪xs:documentation>
      </xs:annotation>
    </xs:attribute>
  </xs:complexType>
</xs:element>
<xs:element name="FCD">
  <xs:annotation>
    <xs:documentation>kontejner pro informace z jednotlivých úseků</
↪xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:group maxOccurs="unbounded" ref="c:SEGMENT"/>
  </xs:complexType>
</xs:element>
<xs:group name="SEGMENT">
  <xs:sequence>
    <xs:element maxOccurs="unbounded" ref="c:SEGMENT"/>
  </xs:sequence>
</xs:group>
<xs:element name="SEGMENT">
  <xs:annotation>
    <xs:documentation>informace pro jeden úsek</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="c:ALL"/>
      <xs:element minOccurs="0" ref="c:CAR"/>
      <xs:element minOccurs="0" ref="c:TRUCK"/>
    </xs:sequence>
    <xs:attribute name="id" use="required" type="xs:NCName">
      <xs:annotation>
        <xs:documentation>identifikátor úseku ze sady předdefinovaných míst</
↪xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="version" use="required" type="xs:integer">
      <xs:annotation>
        <xs:documentation>verze popisu polohy segmentu s daným id</xs:documentation>
      </xs:annotation>
    </xs:attribute>
  </xs:complexType>
</xs:element>

```

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```

</xs:attribute>
<xs:attribute name="last_updated" use="required" type="xs:dateTime">
  <xs:annotation>
    <xs:documentation>čas, pro který jsou data vypočtena</xs:documentation>
  </xs:annotation>
</xs:attribute>
</xs:complexType>
</xs:element>
<xs:element name="ALL">
  <xs:annotation>
    <xs:documentation>hodnoty bez rozlišení typů vozidel (sdružený dopravní proud)</
↪ xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" ref="c:CONGESTION"/>
    </xs:sequence>
    <xs:attribute name="count" use="required" type="xs:integer">
      <xs:annotation>
        <xs:documentation>počet vozidel použitých pro výpočet hodnot</
↪ xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="speed" use="required" type="xs:integer">
      <xs:annotation>
        <xs:documentation>rychlost dopravního proudu [km/h]</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="travel_time" use="required" type="xs:integer">
      <xs:annotation>
        <xs:documentation>dojezdová doba [s]</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="delay" use="required" type="xs:integer">
      <xs:annotation>
        <xs:documentation>zpoždění [s]</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="traffic_level" use="required">
      <xs:annotation>
        <xs:documentation>stupeň dopravy dle zvyklostí v ČR
https://cs.wikipedia.org/wiki/Stupe%C5%88_provozu</xs:documentation>
      </xs:annotation>
    <xs:simpleType>
      <xs:restriction base="xs:token">
        <xs:enumeration value="1">
          <xs:annotation>
            <xs:documentation>stupeň 1: provoz pouze jednotlivých vozidel, jízda
↪ je plynulá</xs:documentation>
          </xs:annotation>
        </xs:enumeration>
        <xs:enumeration value="2">
          <xs:annotation>
            <xs:documentation>stupeň 2: provoz malých skupinek vozidel, jízda je
↪ plynulá, odbavování na křižovatkách bez problémů</xs:documentation>

```

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```

        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="3">
        <xs:annotation>
          <xs:documentation>stupeň 3: tvoří se proudy vozidel, provoz je
↪ plynulý, avšak rychlost nižší než maximální povolená</xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="4">
        <xs:annotation>
          <xs:documentation>stupeň 4: tvoří se kolony vozidel, provoz není
↪ plynulý, průměrná rychlost je výrazně snižena, průjezd křižovatkami je narušen</
↪ xs:documentation>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="5">
        <xs:annotation>
          <xs:documentation>stupeň 5: dopravní kolaps - vozidla na komunikacích
↪ stojí nebo v kolonách jen velmi pomalu popojíždějí, průměrná rychlost je velmi malá
↪ </xs:documentation>
        </xs:annotation>
      </xs:enumeration>
    </xs:restriction>
  </xs:simpleType>
</xs:attribute>
<xs:attribute name="freeflow_speed" use="required" type="xs:integer">
  <xs:annotation>
    <xs:documentation>rychlost volného průjezdu [km/h]</xs:documentation>
  </xs:annotation>
</xs:attribute>
<xs:attribute name="freeflow_time" use="required" type="xs:integer">
  <xs:annotation>
    <xs:documentation>dobu volného průjezdu [s]</xs:documentation>
  </xs:annotation>
</xs:attribute>
<xs:attribute name="reliability" use="required" type="xs:decimal">
  <xs:annotation>
    <xs:documentation>spolehlivost dat, která charakterizuje kvalitu datového
↪ vzorku

```

Nabývá hodnot `<0-1`; kdy 1 znamená 100% spolehlivá data.

Kvalita datového vzorku se hodnotí podle stáří použitých vstupních poloh, jejich
↪ počtu atp. Vzorec výpočtu není součástí tohoto profilu.</xs:documentation>

```

    </xs:annotation>
  </xs:attribute>
</xs:complexType>
</xs:element>
<xs:element name="CAR">
  <xs:annotation>
    <xs:documentation>hodnoty pro osobní vozidla</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" ref="c:CONGESTION"/>
    </xs:sequence>
    <xs:attribute name="count" use="required" type="xs:integer">

```

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```

    <xs:annotation>
      <xs:documentation>počet vozidel použitých pro výpočet hodnot</
↪xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="speed" use="required" type="xs:integer">
    <xs:annotation>
      <xs:documentation>rychlost dopravního proudu [km/h]</xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="freeflow_speed" use="required" type="xs:integer">
    <xs:annotation>
      <xs:documentation>rychlost volného průjezdu [km/h]</xs:documentation>
    </xs:annotation>
  </xs:attribute>
</xs:complexType>
</xs:element>
<xs:element name="TRUCK">
  <xs:annotation>
    <xs:documentation>hodnoty pro nákladní vozidla</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" ref="c:CONGESTION"/>
    </xs:sequence>
    <xs:attribute name="count" use="required" type="xs:integer">
      <xs:annotation>
        <xs:documentation>počet vozidel použitých pro výpočet hodnot</
↪xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="speed" use="required" type="xs:integer">
      <xs:annotation>
        <xs:documentation>rychlost dopravního proudu [km/h]</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="freeflow_speed" use="required" type="xs:integer">
      <xs:annotation>
        <xs:documentation>rychlost volného průjezdu [km/h]</xs:documentation>
      </xs:annotation>
    </xs:attribute>
  </xs:complexType>
</xs:element>
<xs:element name="CONGESTION">
  <xs:annotation>
    <xs:documentation>Popis konkrétní kolony

```

Segmenty jsou interně děleny na menší části (popis jinde).

Na kterékoliv z těchto interních částí vzniká kolona,

pokud na ní rychlost klesne pod 30 % rychlosti volného průjezdu.</xs:documentation>

```

  </xs:annotation>
  <xs:complexType>
    <xs:attribute name="from" use="required" type="xs:integer">
      <xs:annotation>
        <xs:documentation>začátek kolony [m] měřeno od počátku segmentu ve směru
↪dopravního proudu</xs:documentation>

```

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```
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="length" use="required" type="xs:integer">
    <xs:annotation>
      <xs:documentation>délka kolony [m]</xs:documentation>
    </xs:annotation>
  </xs:attribute>
</xs:complexType>
</xs:element>
</xs:schema>
```

Symbols

(W3C) XML schema, [2](#)

D

DDR, [2](#)

F

format, [2](#)

N

NDIC, [2](#)

P

PLS, [2](#)

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R

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U

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